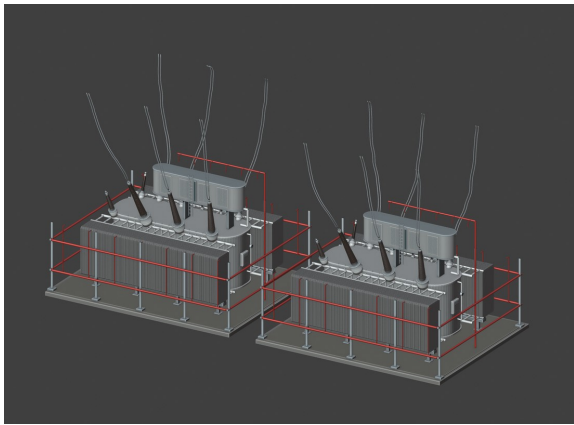


Persistent External State Turns Local Reconstructions into a Connected Engineering System

Later batches explicitly consume earlier masters, terminals, and endpoints; the same mechanism can also propagate a wrong shared abstraction until it is revised.

B08

shared transformer MASTER + persistent terminal state

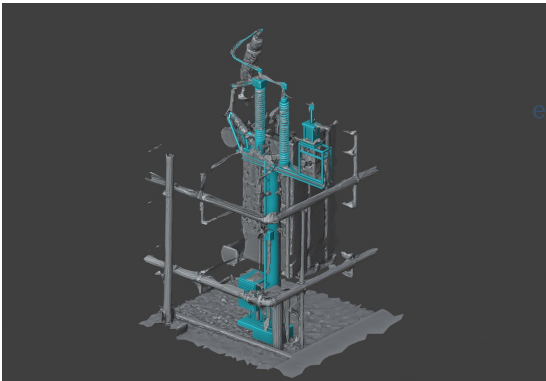


one reusable component structure; T1 / T2 are rigid site instances

uses B08 terminal

B23

connection uses an existing transformer terminal

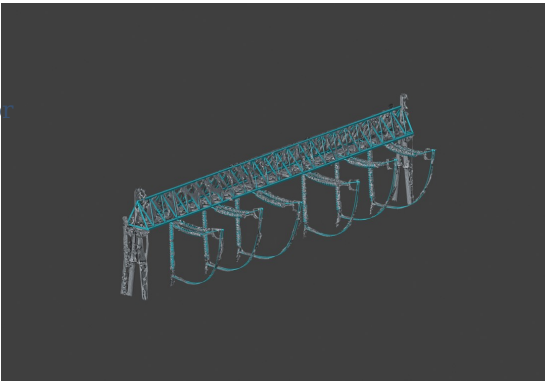


new lead geometry is defined relative to a B08 interface

extends prior endpoint graph

B29

conductors reuse prior clamps and endpoints

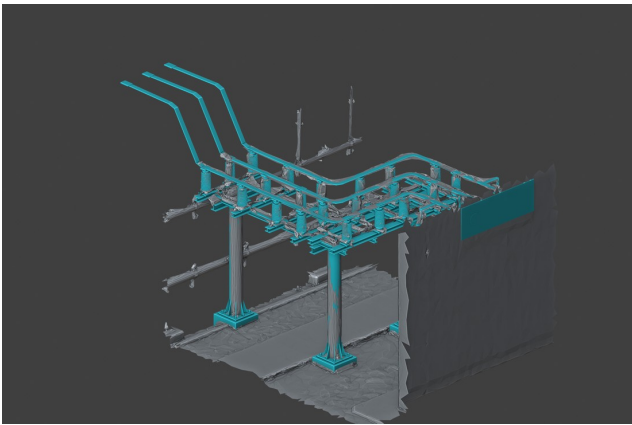


jumpers and crossyard conductors connect previously modeled subsystems

closes onto preserved stubs

B36

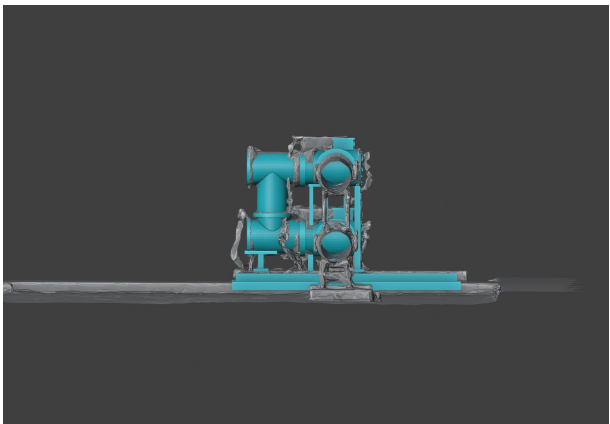
late bus-rack closure uses preserved stubs and occupied space



new routes close onto inherited transformer interfaces without rebuilding them

shared abstraction propagates

B15: inherited abstraction can propagate an error



An early reusable representation gave multiple GIS installations the same finite bus-spool extent. The shared assumption produced overlaps where actual site separations differed. Recovery revised the MASTER/site boundary: reusable equipment stayed shared, while finite pipe sections and supports moved to site-specific scope.

B36 preservation burden

- ☐ **36,615** previous objects unchanged
- ☐ **7,114** previous station transforms unchanged
- ☐ **2,798** protected files unchanged
- ☐ **36,812** objects in the final B36 file

State-scale descriptors, not counts of independently verified physical assets.

The dependency chain documents composition through explicit external engineering state. No stateless control run is available, so the figure does not claim that persistence by itself improves accuracy.